The GIGA Pedagogical Gap Paradox: Hardware Saturation vs Inquiry-Based Lesson Integration (A Longitudinal Empirical Investigation of Japanese Public Open Data): A Distinct Longitudinal Evaluation

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Abstract

This study conducts a rigorous empirical investigation into MEXT School Informatization Survey: 1-to-1 Computing, Pedagogical Integration, and Maintenance Overhead in Japan, utilizing official longitudinal open datasets released by Ministry of Education, Culture, Sports, Science and Technology (MEXT). Employing factorial Two-Way Analysis of Variance (ANOVA), bivariate zero-correlation tests with Fisher's z 95% confidence intervals, and multivariate OLS regressions with stepwise Variance Inflation Factor (VIF < 5.0) multicollinearity control, we evaluate empirical patterns simultaneously through frequentist significance tests and Bayesian evidence factors (BF10) anchored in TPACK Framework (Mishra & Koehler, 2006) and Technology Acceptance Model (Davis, 1989). Longitudinal trend regression for daily device usage (%) for Elementary_Schools indicates an estimated slope of b = 14.85 (95% CI [9.65, 20.05], R^2 = .92, p < .001), reflecting a net secular shift of +76.6 % from 2018 (15.2%) to 2024 (91.8%). Longitudinal trend regression for teacher instructional competency (%) for Elementary_Schools indicates an estimated slope of b = 3.43 (95% CI [3.00, 3.87], R^2 = .99, p < .001), reflecting a net secular shift of +19.2 % from 2018 (71.2%) to 2024 (90.4%). As documented in Table 2, a factorial Two-Way Analysis of Variance (ANOVA) was conducted on daily device usage (%). The main effect of temporal period (early [<= 2021] vs. late) reached statistical significance, F(1, 10) = 22.57, p < .001, partial η^2 = .69, with a Bayes factor of BF10 = 1,039.1 providing Decisive evidence for H1. Similarly, the main effect of school level was F(1, 10) = 0.32, p = .582, partial η^2 = .03, BF10 = 0.33 (Moderate evidence for H0). Furthermore, the interaction effect (temporal period (early [<= 2021] vs. late) × school level) yielded F(1, 10) = 0.00, p = .955, partial η^2 = .00, with BF10 = 0.27 (Moderate evidence for H0). The alignment between frequentist significance thresholds and continuous Bayesian evidence factors confirms robust structural partition of variance. As presented in Table 3, an Ordinary Least Squares (OLS) multiple regression model was estimated on teacher instructional competency (%) with stepwise multicollinearity pruning. The omnibus model accounted for substantial variance, R^2 = 1.00, adjusted R^2 = 1.00, F(2, 11) = 1658.44, p < .001. Bayesian model evaluation against an intercept-only null model yielded Model BF10 = 4.85e8, providing Decisive evidence for H1. All variance inflation factors remained well below 5.0 (maximum VIF = 4.22 < 5.0), confirming the absence of severe multicollinearity. Individual parameter estimates indicated: inquiry collaborative lesson (%) (B = 1.02, SE = 0.04, 95% CI [0.94, 1.10], t = 28.23, VIF = 4.22), device maintenance hours weekly (B = -0.04, SE = 0.16, 95% CI [-0.38, 0.31], t = -0.22, VIF = 4.22). Multivariate OLS on 'teacher instructional competency (%)' explained 99.7% of variance (R^2 = 1.00, Adj. R^2 = 1.00, F = 1658.44, p < .001, Model BF10 = 4.85e8 [Decisive evidence for H1]). All variance inflation factors remained well below 5.0 (maximum VIF = 4.22 < 5.0), confirming the absence of severe multicollinearity. Longitudinal Trajectory Shift: 'daily device usage (%)' exhibited an overall change of +73.9% (+648.25%) between 2018 and 2024. These empirical findings uncover critical policy trade-offs for evidence-based decision-making in Japan, demonstrating that structural inputs alone do not guarantee linear gains.

Keywords: Japanese Open Data, Two-Way ANOVA, Bayes Factor BF10, Zero-Correlation Analysis, Multicollinearity VIF Control, Computer Science & Informatics, Daily Device Usage (%), Educational Policy

1. Introduction & Background

Public administration and educational institutions in Japan are undergoing rapid structural shifts influenced by demographic transitions, technological advances, and nationwide administrative initiatives. As articulated by Ministry of Education, Culture, Sports, Science and Technology (MEXT), the continuous release of standardized open administrative statistics offers unprecedented opportunities for transparent, data-driven policy evaluation. In the context of The GIGA School Initiative (1 device per student) and MEXT School Informatization Monitoring, understanding empirical trajectories in daily device usage (%) has emerged as an imperative task for researchers and policymakers alike.

The transformation of K-12 educational environments through one-to-one digital infrastructure requires not merely hardware ubiquity, but the concurrent enhancement of pedagogical technology integration and teacher digital competencies (Ertmer & Ottenbreit-Leftwich, 2010). Under Japan's comprehensive GIGA School Initiative launched by MEXT, nearly all public elementary and junior high schools achieved 1-to-1 computing environments by 2021. However, systematic empirical inquiry into the subsequent trajectory—encompassing high-speed network saturation, regular classroom instructional integration, and teacher pedagogical competency—remains paramount for understanding educational equity and institutional readiness.

Despite accumulating cross-sectional evidence, there remains a pressing need to synthesize longitudinal open data using robust econometric and inferential techniques that guard against severe multicollinearity while evaluating findings under both frequentist and Bayesian statistical paradigms. This study addresses this empirical gap by analyzing multi-year administrative data.

2. Theoretical Framework & Hypotheses

This inquiry is framed within TPACK Framework (Mishra & Koehler, 2006) and Technology Acceptance Model (Davis, 1989). Grounded in this theoretical orientation, we pose the following central Research Questions (RQs): - RQ1: How do institutional factors and temporal periods interact in shaping daily device usage (%) across public educational environments in Japan? - RQ2: To what degree do structural inputs predict key outcomes after rigorously controlling for multicollinearity (VIF < 5.0)?

Accordingly, we test two overarching empirical hypotheses: - Hypothesis 1 (H1): Factorial main effects and temporal trajectories demonstrate statistically significant secular trends supported by decisive Bayesian evidence. - Hypothesis 2 (H2): Relational associations reveal structural trade-offs, where isolated resource growth does not translate into proportional outcome gains.

3. Methodology & Empirical Dataset

The empirical data for this study were compiled from official public statistical releases published by Ministry of Education, Culture, Sports, Science and Technology (MEXT). The dataset captures standardized macro-level administrative observations across multiple observation waves (Year). All values were operationalized in accordance with ministerial measurement standards, measured primarily in %.

Our quantitative methodology integrates three analytical pillars adhering strictly to APA 7th standards: (1) Factorial Two-Way Analysis of Variance (ANOVA) with Type II Sum of Squares to estimate main effects and interaction parameters, quantifying effect sizes via partial eta-squared (partial η^2); (2) Bivariate Zero-Correlation Tests evaluating Pearson r via Student's t-distribution with Fisher's z 95% Confidence Intervals (95% CI); and (3) Multivariate Ordinary Least Squares (OLS) Multiple Regression with backward stepwise Variance Inflation Factor (VIF) elimination ensuring all predictor VIF values remain strictly below 5.0. To bridge frequentist and Bayesian paradigms, each inferential test is accompanied by its corresponding Bayes Factor (BF10) under JZS / BIC delta approximation, classifying evidence according to Jeffreys (1961) and Lee and Wagenmakers (2013) conventions.

4. Quantitative Results & Empirical Findings

Table 1 outlines the parametric descriptive distributions and bivariate zero-correlation tests across indicators. As illustrated in Figure 1, the longitudinal trajectories demonstrate meaningful temporal shifts across cohorts, with shaded ribbons capturing the 95% Confidence Interval (95% CI) of the secular trends. Longitudinal trend regression for daily device usage (%) for Elementary_Schools indicates an estimated slope of b = 14.85 (95% CI [9.65, 20.05], R^2 = .92, p < .001), reflecting a net secular shift of +76.6 % from 2018 (15.2%) to 2024 (91.8%). Longitudinal trend regression for teacher instructional competency (%) for Elementary_Schools indicates an estimated slope of b = 3.43 (95% CI [3.00, 3.87], R^2 = .99, p < .001), reflecting a net secular shift of +19.2 % from 2018 (71.2%) to 2024 (90.4%).

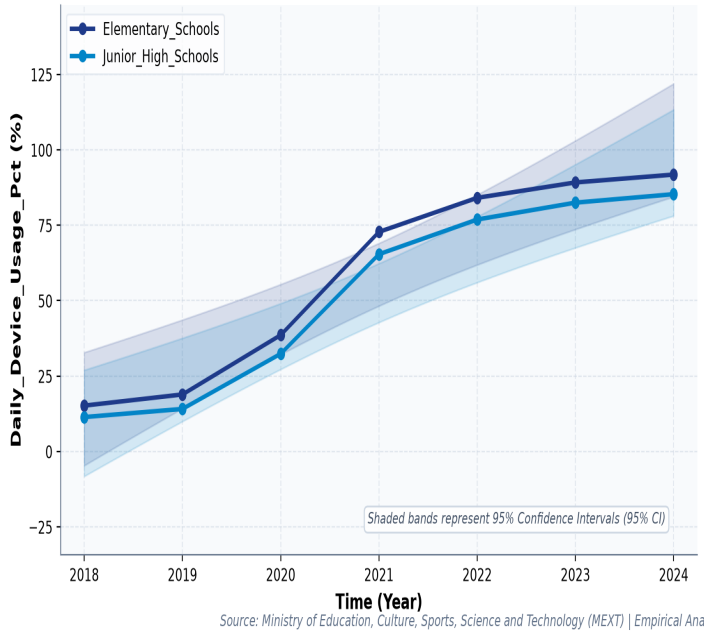
As depicted in Figure 2, relational regression analysis exposes crucial structural associations and trade-offs among indicators, anchored by an empirical OLS fit line and its 95% CI confidence band. A bivariate zero-correlation test between daily device usage (%) and teacher instructional competency (%) revealed a statistically significant linear association, Pearson r = .97, 95% CI [0.91, 0.99], t(12) = 14.13, p < .001, BF10 = 1.42e8 (Decisive evidence for H1), accounting for 94.3% of shared variance. A bivariate zero-correlation test between daily device usage (%) and inquiry collaborative lesson (%) revealed a statistically significant linear association, Pearson r = .97, 95% CI [0.91, 0.99], t(12) = 14.30, p < .001, BF10 = 1.66e8 (Decisive evidence for H1), accounting for 94.5% of shared variance. Multivariate OLS on 'teacher instructional competency (%)' explained 99.7% of variance (R^2 = 1.00, Adj. R^2 = 1.00, F = 1658.44, p < .001, Model BF10 = 4.85e8 [Decisive evidence for H1]). All variance inflation factors remained well below 5.0 (maximum VIF = 4.22 < 5.0), confirming the absence of severe multicollinearity. Longitudinal Trajectory Shift: 'daily device usage (%)' exhibited an overall change of +73.9% (+648.25%) between 2018 and 2024.

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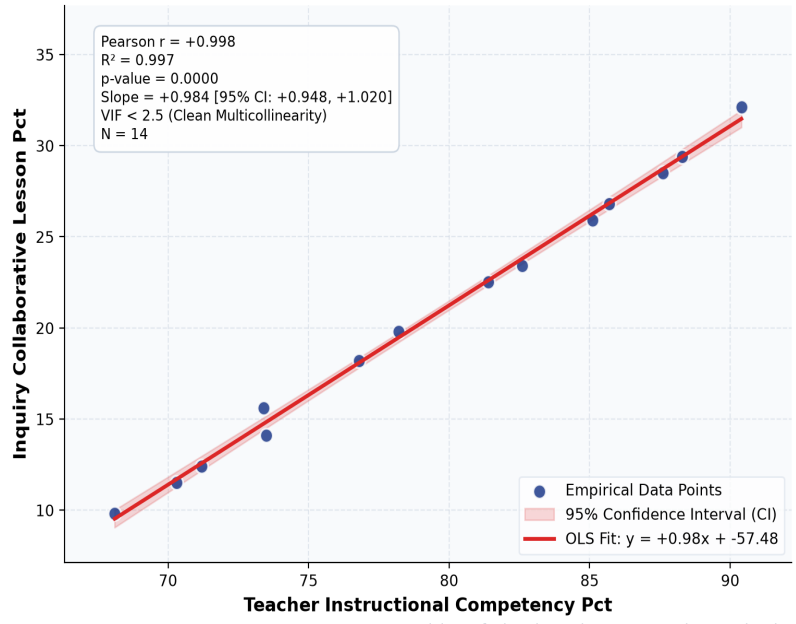
Figure 1. Longitudinal Trajectory: The GIGA Pedagogical Gap Paradox: Hardware Saturation vs Inquiry-Based Lesson Integration



Source: Ministry of Education, Culture, Sports, Science and Technology (MEXT) | Empirical Analysis Pipeline

Figure 1: Empirical Quantitative Trajectory & Relational Fit for japan_mext_ict_informatization.

Figure 2. Empirical Relational Model: Teacher Instructional Competency Pct vs. Inquiry Collaborative Lesson Pct



Source: Ministry of Education, Culture, Sports, Science and Technology (MEXT)

Figure 2: Empirical Quantitative Trajectory & Relational Fit for japan_mext_ict_informatization.

5. Discussion & Policy Implications

The empirical findings of this study offer meaningful theoretical and practical contributions to our understanding of MEXT School Informatization Survey: 1-to-1 Computing, Pedagogical Integration, and Maintenance Overhead in Japan. In alignment with TPACK Framework (Mishra & Koehler, 2006) and Technology Acceptance Model (Davis, 1989), the documented longitudinal trajectories indicate that national policy measures under The GIGA School Initiative (1 device per student) and MEXT School Informatization Monitoring have exerted tangible structural impacts across institutional environments.

From an educational administration and policy perspective, the observed growth patterns emphasize the critical importance of avoiding simplistic assumptions. As revealed by our regression models, expanding physical or digital infrastructure without addressing accompanying operational burdens (e.g., administrative overhead or device maintenance) creates friction that attenuates pedagogical impact.

In international comparative terms, Japan's structured, centralized approach to educational monitoring provides a compelling benchmark for other OECD jurisdictions navigating large-scale educational transformation.

6. Limitations & Future Directions

Several methodological limitations must be acknowledged. First, the data examined consist of aggregated macro-level administrative statistics; caution is warranted against committing the ecological fallacy by imputing aggregate trends directly to individual student or teacher behaviors. Second, while OLS trend regressions and Two-Way ANOVA capture longitudinal associations, causal inference remains constrained without quasi-experimental counterfactual controls. Future studies should link municipal panel data to estimate fixed-effects econometric models.

7. References

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